Wearable IOT System for Real-Time Hand Fatigue Detection During Smartphone Use

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Abstract - **The long usage of the smartphone can lead to the fatigue of the hand, thumb, and wrists because of the repetitive grip force, and incorrect wrists position. The project will describe an IoT based wearable device employed to measure the hand fatigue during the mobile phone use. The proposed system combines a system which operates on the force-sensitive resistors (FSR) sensors on the palm so as to detect the amount of force used to grip an object, and a MPU6500 inertial measurement unit (IMU) to detect the thumb motion and the posture of the wrist. An ESP32 microcontroller constantly reads sensor data, processes it by analyzing it on the microcontroller using a fatigue estimation algorithm which is a set of rules and a simplified algorithm. The system uses sustained grip force, repetitive thumb use and low variation of movement in addition to time in which it is used to determine fatigue risk score in real time. The system also sents a notification message to the user when the fatigue rating exceeds the preset values as a way of persuading the user to tighten the grip, alter wrist- posture or take brief rests. The feedback system will encourage the development of healthier phone usage behavior without visual and audio prompts. All processing is done on the device, and it is possible to achieve a low level of latency response and work without using the cloud services. The suggested system has the capacity to offer an inexpensive, yet efficient means of alerting against the danger of hand fatigue at the earliest stage and introducing ergonomic measures in the long-lasting smartphone utilization by consistent biomechanical-sensing and responsive-tactile-feedback-based control.**

Keywords — Wearable IoT, Hand Fatigue Detection, Sensitive Resistors (FSR), Inertial Measurement Unit (IMU), Embedded Systems

I. Introduction

Hand and finger habits that are now long and repetitive like scrolling, texting and gripping a device have been enabled by the extensive use of smartphones. Such movements require continuous muscular contraction, and they also tend to involve non-neutral locations of the hand and the wrist, which contribute to hand exhaustion and musculoskeletal discomfort. Ergonomic studies have reported that the design properties of smartphones, specifically the size of the screen, play a vital role in the selection of the grip strategy, the accessibility of the thumb, and musculoskeletal load, which are significant to the occurrence of fatigue when using smartphones over prolonged time [9]. In addition to the design of the devices, user behavior has been cited as one of the most significant factors that promote fatigue development. Multiple studies state that there are strong correlations between the changes in hand biomechanics, loss of grip strength, and the more frequent occurrence of hand and wrist pains when using smartphones over a long period of time among university students and young adults [8]. Proper judgment of hand fatigue in actual applications is difficult. Traditional laboratory-based methods such as optical motion capture systems, are highly accurate in measurements, however, cannot be utilized in daily monitoring since the environment is limited, systems are expensive and require a lot of set up. A barrier to dealing with these limitations was overcome by Musso et al. (2025) [1], who have shown that wearable systems that integrate both inertial and force measurements can be used to detect upper-limb movements and grasping behaviors in the field reliably. The wearable sensing technologies particularly Inertial Measurement Unit (IMUs) aid the continuous kinematic monitoring in unconstrained environments.But IMU-only systems do not have any direct indicators of muscle activity and fatigue. To address this limitation, surface electromyography (sEMG) has been widely used to give information about the muscle activities and fatigue development [2]. Although it is effective, sEMG is very sensitive to the location of electrodes, skin preparation and motion artifact, which restricts its use in long-term monitoring of smartphone usage. In order to overcome such issues, force-sensing resistors (FSRs) have become a non-intrusive approach, and previous research has shown that IMU-FSR sensor fusion is a proven detector of grasping behavior and muscle activity under real-world conditions [1]. Recent review works focus on the increased usage of wearable systems and smart data analysis methods to detect a state of continuous fatigue [3]. Though these works demonstrate the developments in wearable technologies and artificial intelligence to detect fatigue, most current systems are developed in occupational or clinical settings and are not aimed at the detection of fatigue related to activities or user intervention in real-time. In spite of the fact that wearable frameworks based on the IoT have been proposed to assess hand functionality [7], the adaptative feedback mechanisms that could be specific to the long-term use of smartphones are still a relatively unexplored field. In order to fill this research gap, this paper is developing an IoT-Based Hand Fatigue Monitoring and Adaptive Feedback System to Prolonged Smartphone Usage through an integration of force and inertial measurability with real-time fatigue detection and user-focused haptic response. The suggested system will

allow constant tracking of hand fatigue when interacting with a smartphone and offer changes in response to fatigue to prevent significant fatigue and encourage healthier interaction habits.

II. RELATED WORKS

In its research, Banadaki et al. (2024) examine how hand pain in university students depends on the patterns of smartphone use. The study in a cross-sectional study of 213 subjects established that extended daily smartphone use had a significant correlation with discomfort, especially of the right thumb, thumb muscle, and wrist. The most common posture that involved holding the phone with right hand and using thumb to use it (41% of students) was most closely linked to pains in particular parts of the body in the hands. They suggest effective interventions: lessening the amount of screen time, using aids, and alternating hands, and these unfavorable consequences and encouraging musculoskeletal health among the frequent users of smartphones [11]. Alenjareghi et al. (2024) present the systematic review of the usage of wearable sensors, such as IMUs, EMG, and pressure sensors, in the prevention of work-related musculoskeletal disorders (WMSDs) as one of the Industry 4.0. The review finds that such technologies can offer real time monitoring and feedback of ergonomics, and thus enhances safety at the workplace and minimizes the risk of injury in such sectors as construction and healthcare. However, some problems like precision of sensors, ease of use, data security and scalability remain. The authors deem the notion of further integration of AI and machine learning in favor of predictive analytics, and the development of standard protocols and ethical systems that would enable more businesses to implement AI and wearable technology to avoid as much as possible [12].

Iadzrola et al. (n.d.) suggest an IoT wearable sensor system as an evaluation of physical effort that combines skin conductance (SC) and electromyography (EMG) signals. By making EMG values the reference to denote low and high level of effort, the method simply classifies effort using normalized SC amplitude through a k-NN algorithm. The isometric arm contractions tested experimentally performed very well with an average classification accuracy of 84.68. The study shows that sensor fusion approach based on SC can be used to measure muscular fatigue effectively and can serve as a substitute to the conventional approaches that rely on acceleration and heart rate [13]. Chen et al. (n.d.) in their study created wearable multimodal sensing system combining fabric based strain sensor, sEMG and a goniometer that monitored fatigue of the biceps brachii during dynamic dumbbell curls. The research results confirmed the muscle thickness to be a legitimate surrogate of sEMG-based activation monitoring and developed a new fatigue biomarker: the difference in maximal activation and peak flexion thickness. The system was able to achieve strong performance with respect to categorizing fatigue states with the use of machine learning models such as SVM and this showed the high effectiveness of multimodal sensor fusion as opposed to the commonly-used sEMG-only instruments in accurately measuring fatigue as well as in detecting transitions in sport science [14].

Gehlot et al. (2022) suggest an IoT-powered wearable that would allow real-time muscle fatigue measurements through the use of a surface electromyography (sEMG) sensor. The system is constructed using an Arduino controller and Wi-Fi module to detect three states of fatigue, which are extensive (>2V), moderate (1-2V), and relaxed (<1V) depending on the average value of RMS of the EMG signal. The data are displayed and recorded using LabVIEW and alerts are presented using color-coded LEDs. It can also send information to a cloud server where it can be analyzed, which illustrates a viable, low-power fatigue-sensing and injury-prevention device in remote athletic and exercise applications [15]. Phadke et al. developed an objective pain and stress-sensing wearable IoT-enabled galvanic skin response (GSR) sensor, which is a wearable device, in their study, including Bluetooth Low Energy (BLE) communication, ergonomic housing, and cloud-based signal processing using the sparsEDA algorithm to filter motion artifacts. In preliminary studies with healthy participants, the prototype was found to be very reliable with regard to detection of GSR alterations caused by stress. This is suggested by the authors as an inexpensive, wireless real-time monitoring tool, with subsequent efforts to be made to miniaturize the hardware and test on the real-world and improve security via the STRIDE framework [16].

Hundekari et al. examined how extended use of smart-phone influenced median motor nerve conduction velocity (NCV) and hand-grip strength in 175 medical students. They reported statistically significant decline in NCV with longer time smartphone use, but not in handgrip strength. The paper postulates that nerve dysfunction and muscle weakness due to repetitive thumb stress and posture strain as a result of overuse of the mobile phone may predispose a person to peripheral neuropathy and other work musculoskeletal diseases [17]. The goal of the wearable IoT system is to evaluate the hand functionality using the EMG of a wristband as proposed by Zhi and Wu to analyze the gesture recognition and grip strength curve regression. In comparison of the four models, they identified that the Random Forest Regression (RFR) model was the best one since it gave a 0.9563 R 2 value to predict grip strength. The paper shows that EMG-based wearable devices are viable in the assessment of hand functions in telehealth through accurate, non-invasive, and continuous methods [18].

The systematical review of wearable-based fatigue monitoring by Adao Martins et al. (2021) notes the application of EEG, PPG, ECG, motion, and other physiological indicators. A majority of the studies utilize supervised machine learning which mostly involves binary classification and record a high reported accuracy but have a limited application to real life as a result of methodological gaps including a short-term laboratory environment, variable fatigue thresholds and poor signal quality measures. The review recommends standardized procedures, real-world validations in the long term, and better data quality as the means of increasing the reliability and generalizability of wearable fatigue monitoring platforms [19]. In Zhou et al. (2024), the author constructs a data glove on the basis of a wearable technology (four 6 axis IMU sensors) to measure hand functions and fatigue accurately. The instrument has a

good measurement of the finger joint kinematics, which have been proven by comparing the angles of the instrument with the static and dynamic angles. Gripping experiments measure the range of motion of the joints and finger tapping tests show fatigue by decreasing the angular velocity frequency and amplitude with time. The system also provides a portable objective alternative to subjective scales and expensive motion capture systems that have shown potential to provide personalized rehabilitation monitoring and real time fatigue tracking during hand therapy [20].Kakhia, K. et al. (2025) carried out a critical review of wearable devices and AI to recognize fatigue through physiological measures, including ECG, EMG, and EEG.It emphasizes the fact that fatigue can be better assessed using multimodal signal analysis and ML/DL models. The paper recognizes the issues of real-time access of data, ergonomics and explainable AI require and suggest future work directions, such as edge computing and advanced sensor fusion [21].Karim, Guo, and Wu (2025) survey the development of the construction worker fatigue (CWF) analysis, and combine the research on physical fatigue and mental fatigue with the new technologies, such as wearable and artificial intelligence (AI) and computer vision. They mention the tendencies in multimodal sensing and machine learning to detect fatigue in real-time and promote the use of more personalized, context-sensitive and culturally sensitive fatigue management systems in the future to enhance construction sites safety and productivity [22].

In their study, Musso, Oliveira, and Bai (2024) experimentally assess the influence of a passive exoskeleton on the upper limb on muscle activity in conducting overhead assembly, bricklaying, and box-moving tasks. The exoskeleton highly minimised the activation of the shoulder flexors (maximum -45.46% anterior deltoid) of all tasks but had an augmented extensor activation when working on the sub-shoulder. Findings indicate the effectiveness of the device in overhead support and its performance in regard to tasks, which means that posture-specific design and use are needed to maximize fatigue benefits and eliminate work-related musculoskeletal disorders in construction and manufacturing [23].Chen et al. (2026) provide the review of wearable inertial measurement units (IMUs), insole pressure sensors (IPS), and surface electromyography (sEMG) usage in biomechanical monitoring of exercise fatigue. The article summarizes the development in the real-time detection and machine-learning components by summarizing the advances in the understanding of fatigue-induced modification of joint kinematics, ground reaction forces, and muscle activity using these devices. It determines the main obstacles, such as signal noise, device longevity and requirement of multimodal fusion and future prospects of creating robust, personalized and continuous fatigue-management systems to increase athletic performance and workplace safety [24].

Sarvghadi et al. (2025) survey wearable technologies to measure five human factors trust, cognitive workload, stress, perception of safety and fatigue- in human-robot collaboration. Their survey singles out popular gadgets and physiological metrics (EDA, HRV, EEG). It has been found that the most researched factors are trust and cognitive workload, and the challenges mentioned in the analysis, such as the interoperability of the device, ergonomic design, and real-time processing. The future will see the addition of the AI and standardized sensor systems to increase HRC effectiveness [25]. Ajayi et al. (2025) guide through the haptic technologies on hardware-in-the-loop development i.e., sensors, actuators, and materials. The four types of haptic feedback vibrotactile, tactile, force, and thermal are analyzed in the paper and the materials analyzed. Applications of interest in manufacturing, mining, medical and training are identified. The problems that are discovered during the review include low localization, slow thermal response, bulkiness, and latency. Future perspectives are multimodal sensing, better actuator design, AI integration, and smaller and lower-power systems to increase haptic realism and usability in immersive and remote operating conditions [26].

Tang et al. (2024) overview the development of the haptic interface as a part of the immersive metaverse experience. They point out that the main issues with the haptic technology are rendering realism, latency, and reproduction of textures, power supply, and cost to fabricate the device. The future opportunities entail the use of 6G to reduce latency and improve realism by employing integrated haptic stimuli. The authors point out the necessity of consistent assessment systems to faster the creation of market-ready, haptic systems of immersions in VR/AR applications [27].Simatic and Stojanovic (2024) designed a two-channel 12-bit wearable EMG feedback device, which is connected to Bluetooth and equipped with real-time cross-correlation algorithm in-situ to analyze muscle activity. It is economical, portable and allows personalized rehabilitation with a good track record of reliability in validating the system with volunteers [28]. Jiang et al. (2024) reviewed the literature on the self-powered wearable internet of things sensors in human-machines interfaces and examined 113 journal articles. They established four major areas of research, namely TENG/PENG energy harvesting, flexible electronics, Industry 4.0 integration, and HMI devices. The review identifies existing gaps in the research and future trends in the evolution of wearable sensors in construction and other areas [29]. Ahamed et al. (2024) suggested an Energy-Efficient Healthcare Data Management Method (EE-HDMM) based on the IoT-assisted wearable sensors and cloud-based health monitoring. This paper presents an IoT-based wearable sensor platform of real-time health monitoring with focus on energy efficiency, low-latency data transmission, and cloud-assisted processing [30].

III. PROPOSED METHODOLOGY

3.1. System Overview

The suggested solution is an IoT smart glove that is a wearable to control the fatigue of hand muscles during the extended use of a smartphone. The constant holding of the phone, scrolling of thumbs, and placing the wrist in a static position results in muscle stress and exhaustion. The system measures the fatigue in real-time based on the force and motion sensing and promptly gives a notification to the user.

The device is completely powered by an embedded processor and needs very little external computation. Grip force and wrist/thumb movements sensor data are constantly gathered, transformed, and processed to produce a fatigue score. Once this score is above a given limit, the user receives a wireless

notification via Telegram alerts. The entire processing chain is split into three significant functional modules Sensing and Data Acquisition, Signal Processing and Fatigue Estimation and Feedback Communication. All these modules allow continuous monitoring and intervention in time as shown in system architecture in figure 1.

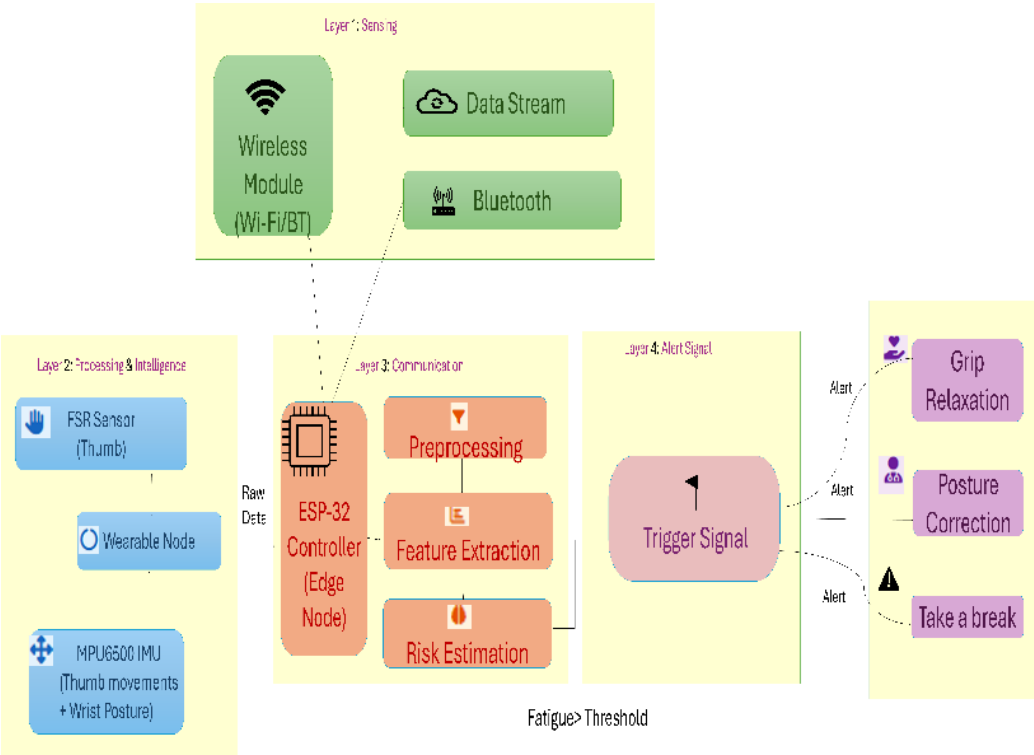

Fig 1. Proposed System architecture to detect hand fatigue during smartphone use

3.2. Sensing and Data Acquisition in Wearable

The module is the basis of the suggested hand-fatigue monitoring system as it allows monitoring of physiological and biomechanical signals in the hand of the user continuously. Given that estimation of fatigue is solely reliant on the accuracy and reliability of the sensed data, it is important that wearable glove sensors be placed and integrated carefully. The design of the glove will make sure that grip force, the position of the wrist and the movement of the thumb is registered in real time when using the smartphone normally without limiting the movement of the user. The signals that are obtained are sent to an embedded processing unit to analyze them further. Through direct on-body sensing, the system avoids the use of external cameras or other cumbersome equipment, thus being more portable and comfortable and practical to be utilized in day-to-day life.

3.2.1 Embedded Processing Unit

The ESP32 is the main processing unit of the system that serves as the brain of the wearable device. This microcontroller is in charge of communicating with every sensor, converting the analogue signals into a digital one with its in-built Analog-to-Digital Converter (ADC), and communicating with inertial sensors through its I2C protocol. In addition, it calculates real-time features, the level of fatigue, and sends warning messages via wireless communication. The main reason to consider the ESP32 is the ability to incorporate the WiFi functionality into the chip, which consequently eliminates the use of other communication devices. The reason is its low power use that can be used in battery-powered wearable devices, and its multi-channel ADCs enable parallel attachment of force sensors. The adequate processing speed and memory limit allow on-server computing without the use of external servers. Having all the processes in a single location minimizes the system lag, makes them respond better and increases the system reliability even during unstable internet connections.

3.2.2. Grip Force Sensing Using FSR

The measurement of the grip force is done by using a Force Sensitive Resistor (FSR) placed at the palm of the glove. The FSR working principle is founded on the negative correlation between the applied force and electrical resistance. The greater the grip pressure on a smartphone when the user is holding a phone, the lower the resistance of the sensor. This variation in resistance is then wired in a voltage divider arrangement to produce a voltage that is measurable. The resultant output voltage is as follows is shown in equation 1:

$$V_{out} = V_{cc} \times \frac{R_{fixed}}{R_{fixed} + R_{FSR}} \qquad (1)$$

where R_{FSR} is the resistive variable of the sensor. Since the applied force is inversely proportional to the sensor resistance, the equation 2 represents the force as

$$F \propto \frac{1}{R_{FSR}} \qquad (2)$$

then, it will be true that the tighter the grip, the more the output voltage. Constant gripping leads to permanent contraction of the muscle thus limiting blood flow and causing muscular fatigue. Thus, grip force is a critical measure of physical load and is part of the general fatigue measure.

3.2.3. Improved Motion and Posture Sensing with IMU

It also has two MPU6500 inertial measurement units to monitor the movement of the wrist and thumb in addition to force sensing. In every sensor, there is a three-axis accelerator and three-axis gyroscopes, which allow the measurement of both the linear and angular velocity. These are measurements that contain detailed information on the overall posture of the wristnand the repetitive movements of the fingers. The gyroscope outputs are of great use especially in determining the scrolling, typing, and tapping which are repetitive actions of a smartphone interaction. On the same note, long-term, static, and wrist position can also be observed in cases where angular velocity tends to go near the zero state during longer periods of time. Repeated movement and prolonged holding are the two factors that have been established to cause ergonomic stress and muscular fatigue. Through continuous capture of such patterns of motion the system can also know about the behavioral patterns of usage by the user which are subsequently used in the process of estimating fatigue.

3.3 Signal Processing and Fatigue Estimation

The raw measure of the sensors usually has errors in the measurements, environmental signals and discontinuous variations, which can result into wrong prediction of fatigue. Thus, this module is dedicated to the optimization of the obtained signals with the use of filtering and feature extraction methods and the calculation of a quantitative fatigue measure. The problem is to convert unstable sensor values into consistent and significant parameters that provide a true picture of the physiological state of the user.

3.3.1 Noise Filtering

An exponential low-pass filtering method is used to deem undesired noise and even out the sensor signals coming in. In this way, sharp rises due to vibrations, electric interference or other uncommon motions are eliminated, but the general direction of the signal is retained. The mathematical expression of the filtering operation is given in equation 3 as:

$$y[n] = \alpha x[n] + (1-\alpha)y[n-1] \qquad (3)$$

where the input sample of the raw data is denoted x[n], the output of the filter is denoted by y[n] and the smoothing factor is denoted ais in the range of 0 to 1. Lower value of a will give smoother signals but with lower response and higher value will enable faster adjustment to changes. This step of filtering guarantees the stable measurements and reduces false fatigue alarm rate.

3.3.2 Feature Extraction

Following the filtering, the system identifies some of the statistical features describing the intensity and pattern of hand activity. The average grip force over a time window is computed in equation 4 as:

$$F_{mean} = \frac{1}{N}\sum_{i=1}^{N} F_i \qquad (4)$$

that is the total muscular weight load of the user.

Repetitive thumb movements such as scrolling and typing are identified from the gyroscope signal. The repetition rate is computed in equation 5 as:

$$R = \frac{N_{peaks}}{T_w} \qquad (5)$$

Where N_{peaks} = number of gyroscope peaks detected in window T_w and T_w= time window duration (seconds).

Additionally, cumulative usage time is tracked in equation 6 as:

$$T_{usage} = \sum T_w \qquad (6)$$

where T_w stands as windows of active time. The length of time before a rest is a major cause of strain on muscles.

Motion variability is calculated to measure the degree of wrist movement and is defined in equation 7 as:

$$Var_{motion} = \frac{1}{N}\sum(\omega_i - \bar{\omega})^2 \qquad (7)$$

where ω_i is the angular velocity sample and $\bar{\omega}$ is its mean. Low variability means that it is in a static position, exposing it to fatigue. Combined, these characteristics present a full picture of force, movement and time-related stressors.

3.3.3 Fatigue Score Model

In order to obtain the general fatigue status, a lightweight rule-based fatigue score is produced by adding all the extracted features. The fatigue score is expressed in equation 8 as:

$$S = w_1 F_{mean} + w_2 R + w_3 T_{usage} + w_4\left(\frac{1}{Var_{motion}}\right) \qquad (8)$$

where $w_1, w_2, w_3,$ and w_4 are weighting coefficients. When the computed score exceeds a predefined threshold, it can calculated in equation 9 as:

$$S > S_{th} \qquad (9)$$

the system classifies the condition as fatigue. The use of rule-based over more complex machine learning models is due to the fact that rule-based requires less memory, computation is faster, and the process of training is non-existent, thus making it extremely appropriate in real-time embedded applications.

Algorithm Real-Time Hand Fatigue Detection and Feedback

Input: FSR sensor data (grip force), IMU data (acceleration and angular velocity)
Output: Fatigue warning message sent through Telegram
Initialize:
fatigue score S
usage time T_{usage}
fatigue threshold S_{th}
filter coefficient α

```
while device power is ON do
    Acquire raw FSR value F_raw
    Acquire IMU acceleration and gyroscope values
    Apply low-pass filtering
        y[n] = αx[n] + (1 − α)y[n − 1]
```

Normalize filtered signals
Compute features
$F_{mean} \leftarrow$ average grip force
$R \leftarrow$ repetition rate
$T_{usage} \leftarrow$ cumulative usage time
$Var \leftarrow$ motion variability
Calculate fatigue score

$$S = w_1 F_{mean} + w_2 R + w_3 T_{usage} + w_4 \left(\frac{1}{Var_{motion}}\right)$$

if $S > S_{th}$ **then**
Send Telegram alert message
else
Continue monitoring
end while

3.4 Feedback Communication

This module will have the task of giving timely feedback to the user after fatigue has been detected. Early intervention is required to avoid prolonged strains and enhance healthy habits of using the devices. Once the score of fatigue reaches the limit, the system sends a warning message and triggers an alert system. The warning prompts the user to loosen their grip, change a position of the wrist or make brief pauses to rest. To enable remote notification, ESP32 uses WiFi capabilities incorporated in the device to send alert messages using the Telegram application programming interface. Using the Telegram bot service, it is possible to send the HTTP requests and notify the user of his/her smartphone instantly. This methodology facilitates real-time capability, mobility, and portability without the need to use extra hardware units.

IV. RESULTS AND DISCUSSION

4.1 Experimental Results

The prototype shown in figure 2 was put through the various conditions of smartphone use, such as normal grip, sustained static holding, repetitive thumb scroll, and sustained non-neutral posture of the wrist. The palm-held FSR sensor had more grip force on prolonged holding, which was a sign of static muscular loading. The wrist MPU6500 detected well the posture measurements on moving off the neutral position, whereas the thumb-mounted MPU6500 recorded the repetitive motion pattern of scrolling.

Fig 2. Prototype for hand fatigue detection

The calculated fatigue score rose steadily as high grip force, relative to persistence of the wrist deviation, and greater repetition of the thumb occurred jointly with the continuous time of use. In the usual usage, the fatigue score had not exceeded the predetermined value and the system state was kept as the one that was defined as the value of "OK" . Under the conditions of combined stress, the fatigue score was over the threshold and a Telegram alert notification was sent successfully as seen in figure 3. The system was found to be reliable with a USB power supply, as it responded to the alert in less than one second and no false alarms during relaxed conditions.

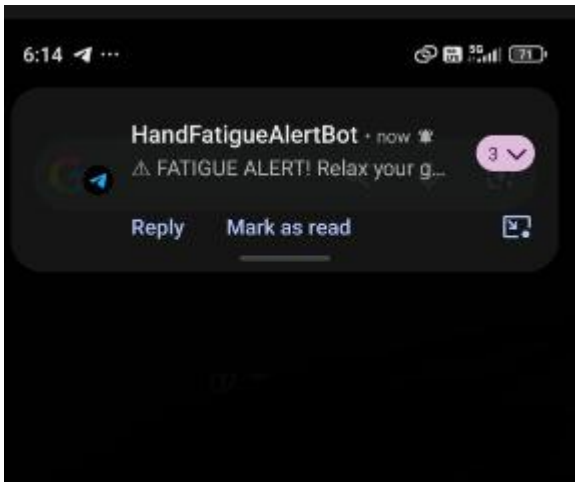

Fig 3. Alert message notification using Telegram bot service

4.2 Discussion

The experimental results demonstrate that combining pressure sensing with dual IMU-based motion tracking improves the reliability of hand fatigue recognition during prolonged smartphone usage. The FSR sensor measures excessive grip force applied to the smartphone, while the wrist IMU detects posture deviations and the thumb IMU captures repetitive thumb movements. The graphical results obtained from grip force variation, wrist motion patterns,

thumb movement activity and fatigue score progression further support the effectiveness of the proposed multi-sensor approach. Compared with single-sensor methods, this multi-sensor fusion technique provides more robust and accurate detection of fatigue-related behaviours.

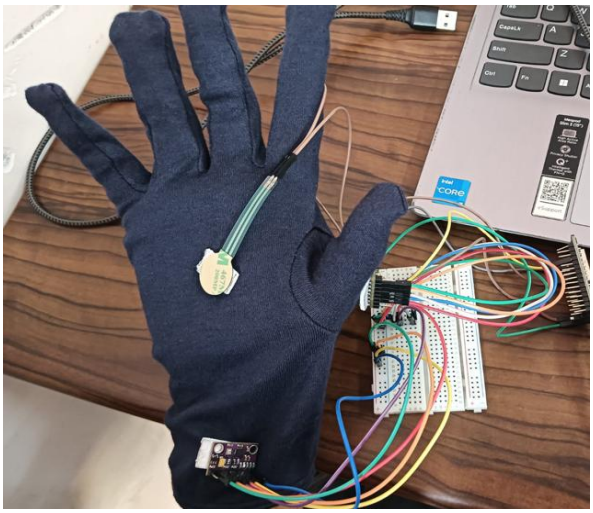

Fig 4. Wearable prototype with FSR and IMU sensors used for smartphone hand fatigue monitoring

The figure 4 shows the mbedded IoT wearable glove designed to detect hand fatigue during prolonged smartphone use. An ESP32 microcontroller acquires data from the FSR grip sensor and MPU600 motion sensors through ADC and I²C interfaces. The embedded algorithm processes signals locally to compute fatigue levels and triggers real-time Telegram alerts.

The real-time operation testing of the embedded wearable system when using a smartphone is shown in figure 5. The FSR sensor will read the changes in the grip force, and the MPU6050 IMU will record the wrist orientation and thumb movement. When sensor data sent to ESP32 microcontroller where embedded firmware analyzes signals, calculates fatigue parameters, and sends wireless Telegram alerts via Wi-Fi connection when set predetermined thresholds are exceeded.

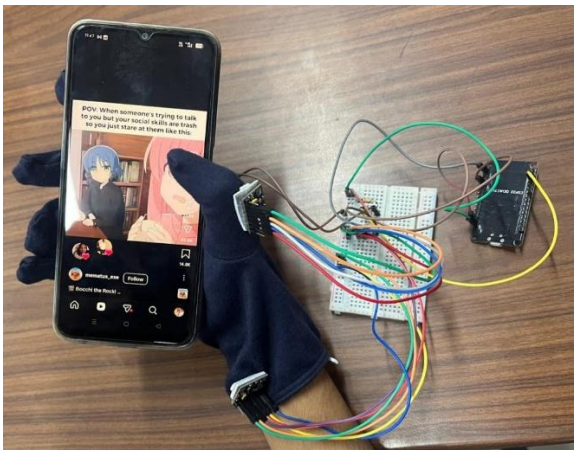

Fig 5. Experimental setup showing glove-mounted sensors used to monitor grip force and hand motion during smartphone usage

ESP32 assists in low latency computation and eliminating cloud infrastructure dependency. The alert mechanism is a Telegram-based mechanism that offers a simple and scalable feedback mechanism. Comprehensively, the system confirms the practicability of a low-cost, real-time biomechanical fatigue monitoring system to the long-term use of a smartphone.

Sensor	Parameter Measured	Converted Sensor Range	Threshold Condition	Fatigue Score Weightage
FSR (Force sensitive resistor)	Grip force	0 – 4095 (ESP32 ADC)	Grip > 200 ADC	+30
Thumb MPU6500	Thumb motion variation	0 to +250°/s	ThumbVar > 20°/s	+30
Wrist MPU6500	Wrist motion variation	0 to +250°/s	WristVar < 2°/s	+20

Table 1. Sensor Threshold and Fatigue Scoring Parameters

The table 1 summarizes the sensors used in the fatigue monitoring system, the parameters they measure, their operational ranges, and threshold conditions. Each sensor contributes a weighted score based on detected grip force, thumb movement, and wrist stability to estimate overall user fatigue.

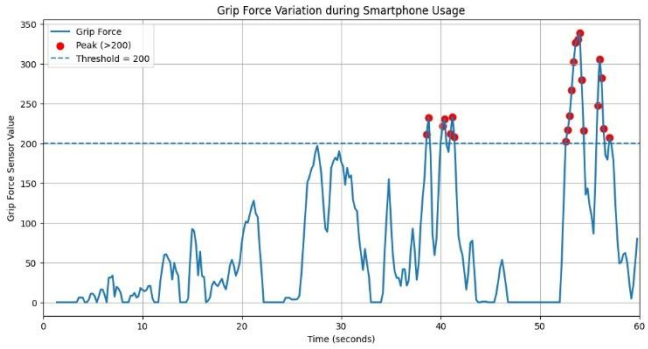

Fig 6. Grip Force Variation Measured Using Force-Sensitive Resistor (FSR) During Smartphone Usage

The graph in figure 6 indicates the real-time grip pressure unit on the palm by the FSR sensor. The signal is the tightness of the grip of the smartphone to the user as time passes. Mountains above the threshold reflect perpetual or excessive grip force, which assists in straining and wearying the muscles. Repeated high pressure activities are indicative of constant stress on hand muscles. This parameter is one of the key biomechanical parameters that the ESP32 system uses to identify the prolonged use of force and approximate fatigue risk.

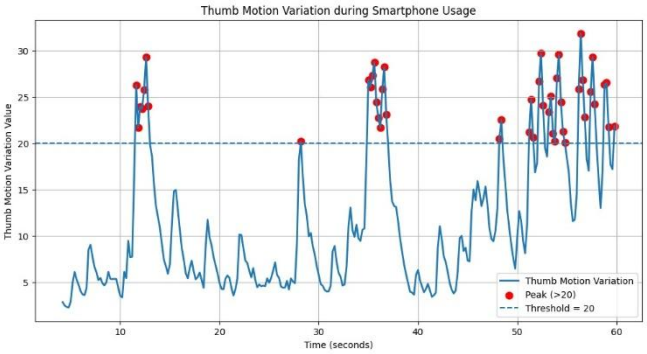

Fig 7. Thumb Motion Variation for Repetitive Movement Detection

The plot in figure 7 corresponds to the intensive movement of the thumb, which was recorded with the MPU6500 accelerator attached to the thumb. Movement change is an indication of scrolling, typing, and gestures when using the phone. Thumb motion with spikes above the threshold is a repetitive and high-frequency movement that is a major cause of tendon strain and overuse fatigue. Sustained peaks imply intensive interaction patterns. This variability of motion is analyzed by the system to measure repetitive hand activity which is a significant consideration in real time estimation of ergonomic fatigue.

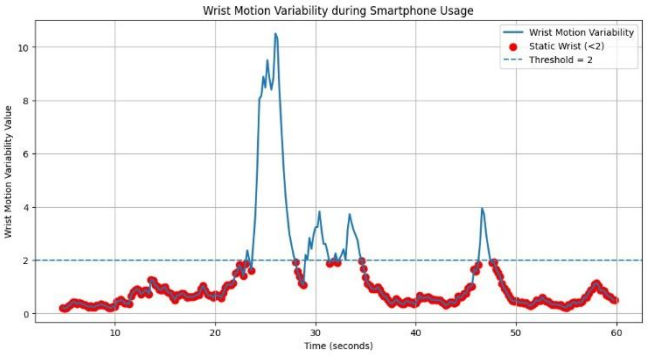

Fig 8. Wrist Motion Variability Analysis During Smartphone Interaction

The variability of wrist movement recorded on the palm-mounted MPU6500 is displayed in this fig.8. The lower values of variability show a steady posture of the wrists whereas the higher values are dynamic movement. Extended positions whose threshold is below the threshold indicates permanent fixed in the position of the wrist which may decrease blood circulation and raise musculoskeletal strain. Posture and grip force in a static posture increases the risk of fatigue. The wearable device keeps an eye on the variability of the wrist to identify ergonomic strains due to limited movement during long periods of smartphone usage. There is sudden spike in the graph which may indicate sudden change in wrist posture.

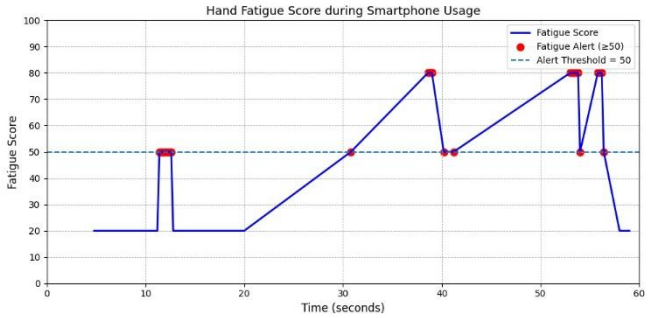

Fig 9. Hand Fatigue Score Estimation During Continuous Smartphone Usage

The figure 9 shows the calculated fatigue score produced by the ESP32 with a rule-based model. The score combines grip force, repetition of thumb motion, variation of wrist movements as well as duration of use. Values beyond the alert threshold are an indication of high risk of fatigue that needs to be interfered with by the user. Peak levels are associated with sustained grip and repetitive movements of limited wrist movement. This cumulative measure allows real-time fatigue measurement and initiates alert system.

CONCLUSION

This paper presented a wearable IoT-based system for real-time detection of hand fatigue during prolonged smartphone usage. The proposed device integrates a palm-mounted FSR sensor to measure grip force and an MPU6050 IMU sensor to monitor wrist posture and thumb motion. Sensor data were processed locally using an ESP32 microcontroller, where a fatigue estimation model calculated a fatigue score based on sustained grip force, repetitive thumb movements, and non-neutral wrist positions. The experimental results, illustrated through graphical analysis of grip force variation (Fig. 6), thumb movement patterns (Fig. 7), wrist motion variability (Fig. 8), and the overall fatigue score progression (Fig. 9), demonstrate the capability of the system to effectively detect fatigue-related conditions during extended smartphone use.

All computations were performed on-device, ensuring low latency and eliminating dependence on cloud services. The prototype operated reliably under USB power and maintained stable wireless communication while sending real-time alerts through Telegram whenever the calculated fatigue score exceeded a predefined threshold. The combination of force sensing and inertial motion monitoring improved the reliability of fatigue detection compared to single-sensor approaches, while the graphs provided clear visualization of how sustained strain conditions gradually increase the fatigue score.

Prolonged smartphone usage with excessive grip force, repetitive thumb movement and improper wrist posture can increase the risk of musculoskeletal problems such as repetitive strain injury (RSI), tendon inflammation, wrist strain and early symptoms of carpal tunnel syndrome. By continuously monitoring these parameters, the proposed system can help users recognize unhealthy usage patterns and encourage timely breaks, thereby reducing long-term health risks. Overall, the system provides a low-cost, portable and scalable solution for early hand fatigue detection. Future improvements may include rechargeable battery integration,

micro-vibration haptic feedback, miniaturized hardware design and adaptive fatigue models for personalized health monitoring.

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